OPTIMIZING SOIL AMENDMENTS VIA PRECISION FERTILIZER APPLICATION:

A COMPARATIVE STUDY OF ENSEMBLE MACHINE LEARNING MODELS

Memory Chibwe, Pilirani Mogha, Everson Hara

Department of Agricultural and Applied Economics

February 1, 2026

Introduction

- Background & Motivation
- Problem Statement
- Objectives
- Dataset Overview
- Methodology & Tools
- Results and Conclusion

Background & Motivation

- Fertilizer application is critical for agricultural productivity.
- Improper use leads to soil degradation and low crop yields.
- Machine learning enables precision fertilizer recommendations.

Problem Statement

- Uniform fertilizer application ignores soil variability.
- Incorrect fertilizer choice increases cost and environmental risk.
- Need for intelligent, data-driven decision support.

Objectives

- Develop a machine learning-based fertilizer recommendation model.
- Compare Random Forest and XGBoost algorithms.
- Identify key soil and environmental factors.

Dataset Overview

- Dataset: `fertilizer_recommendation_dataset.csv`
- Supervised multi-class classification problem.
- Soil, environmental, and crop attributes.

Distribution of Fertilizer Recommendations

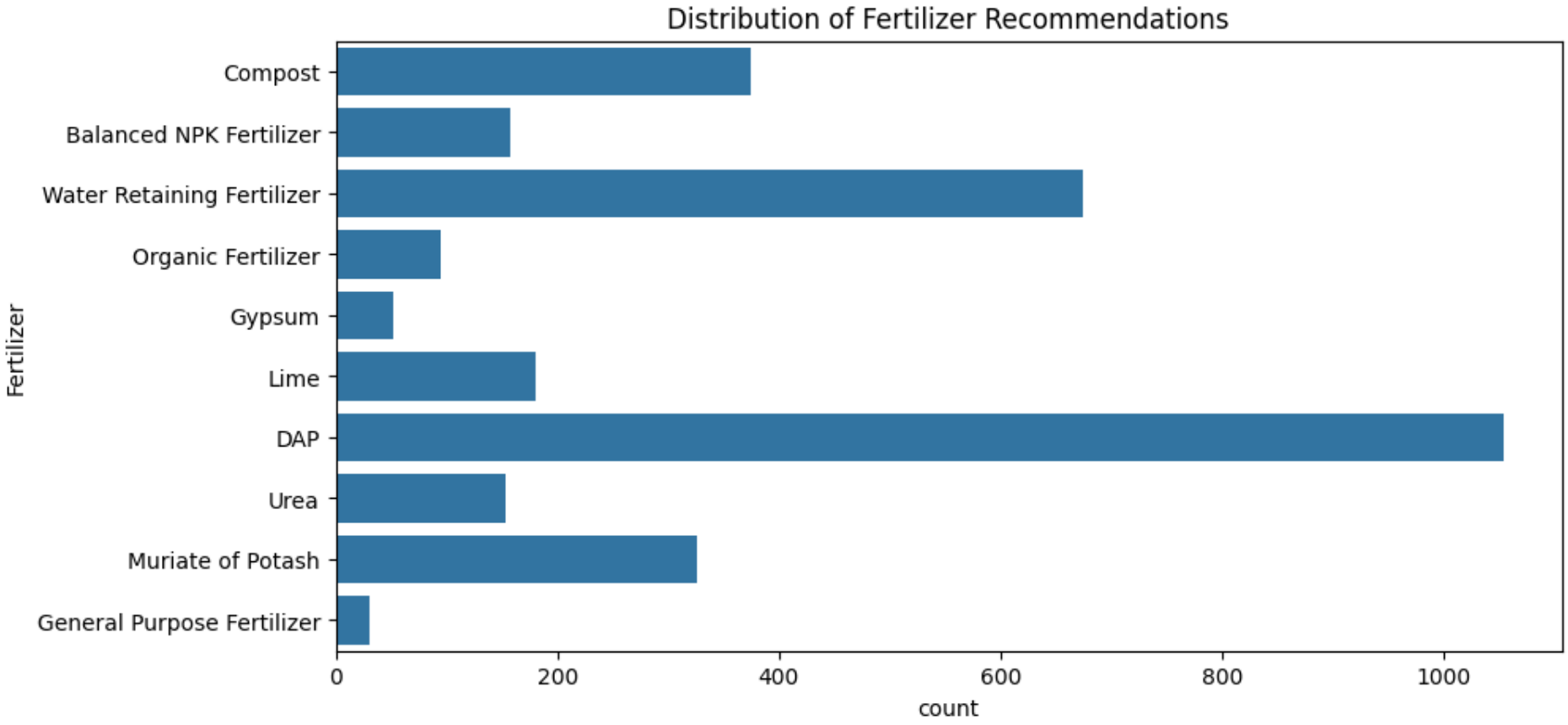


Figure 1: Distribution of fertilizer recommendations

Why Python?

- Efficient handling of structured datasets.
- Reproducible and transparent workflows.
- Rich ecosystem for machine learning and visualization.

Tools & Libraries

- Pandas, NumPy
- Scikit-learn, XGBoost
- Matplotlib, Seaborn
- SHAP (SHapley Additive exPlanations)

Data Preprocessing

- No missing values detected.
- Numerical variables standardized.
- Categorical variables One-Hot encoded.
- Train-test split (80% / 20%).

Correlation Analysis of Numerical Features

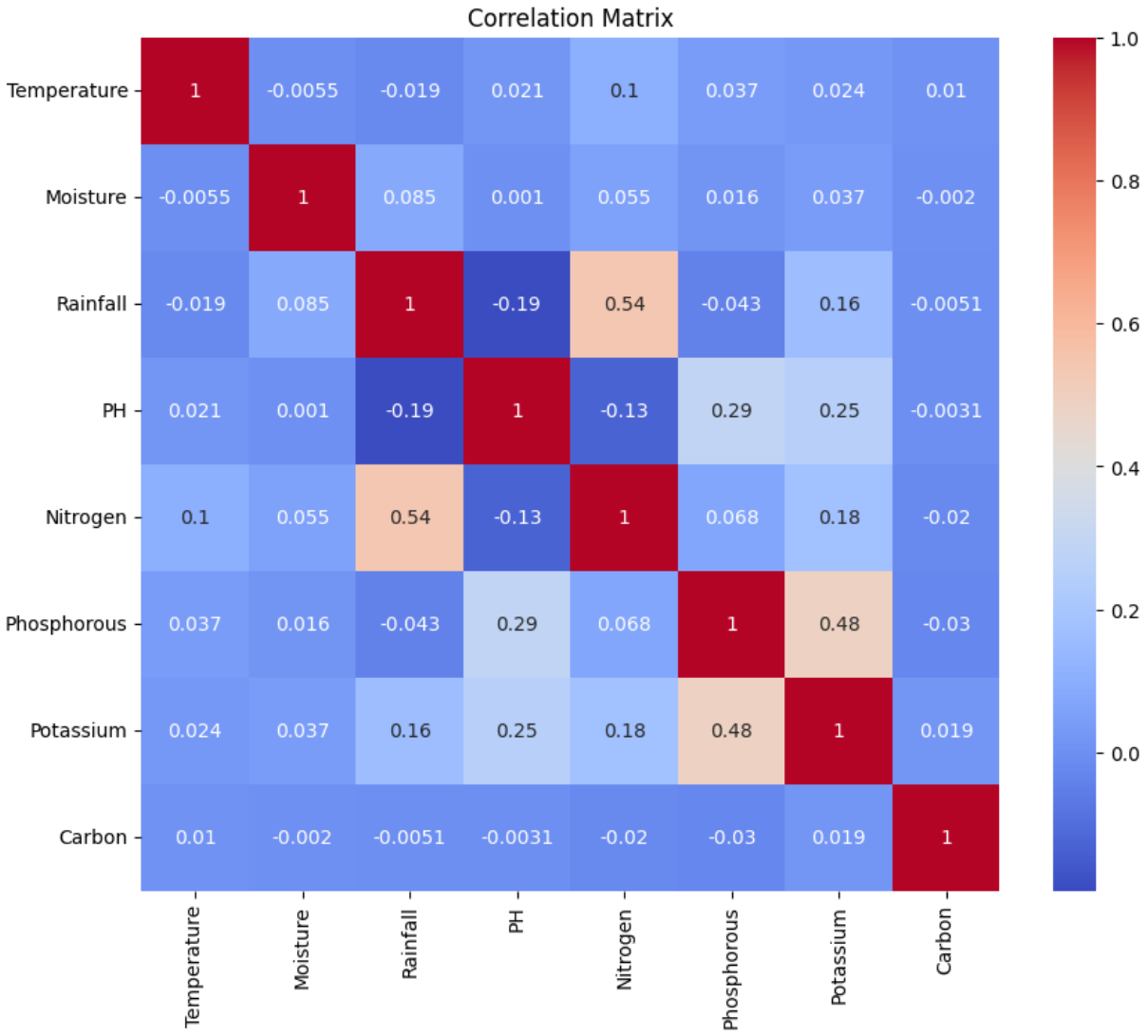
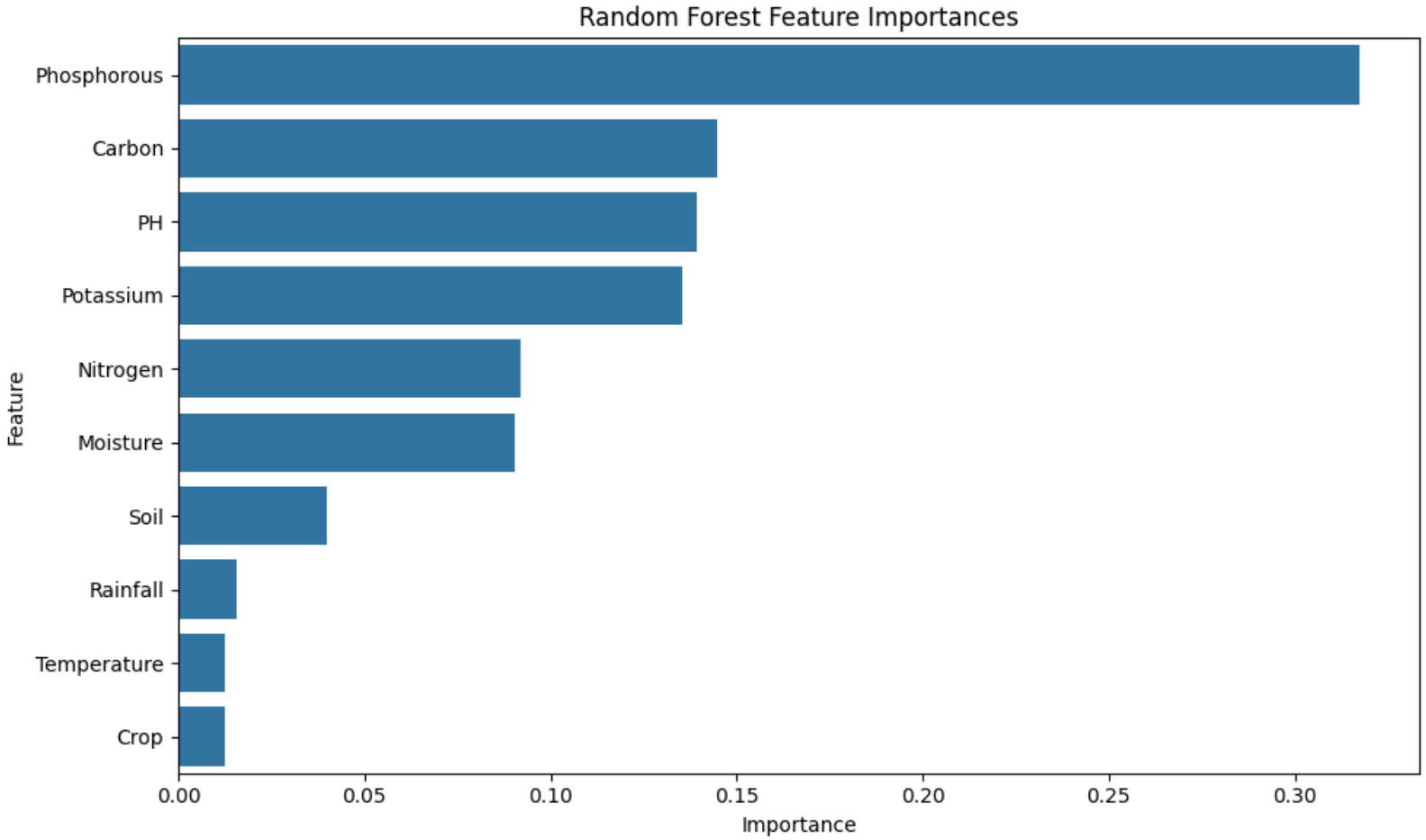


Figure 2: Correlation heatmap showing relationships among numerical features

Analytical Approach

- Supervised classification.
- Random Forest and XGBoost models.
- Evaluation using accuracy and confusion matrix.
- Model explainability using SHAP.

Feature Importance (Random Forest)



Confusion Matrix – XGBoost

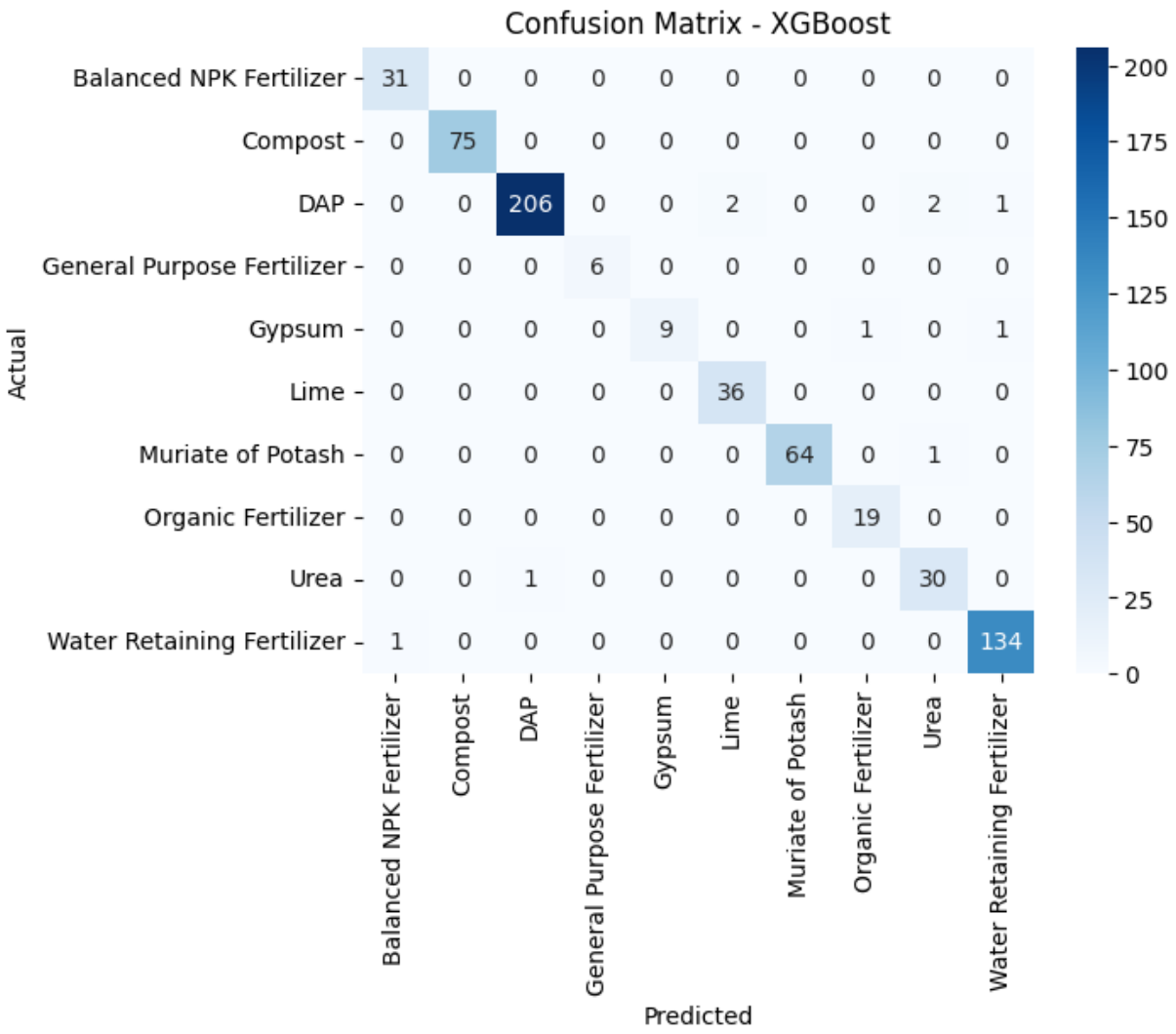


Figure 3: Confusion matrix for the XGBoost classifier

Model Explainability using SHAP

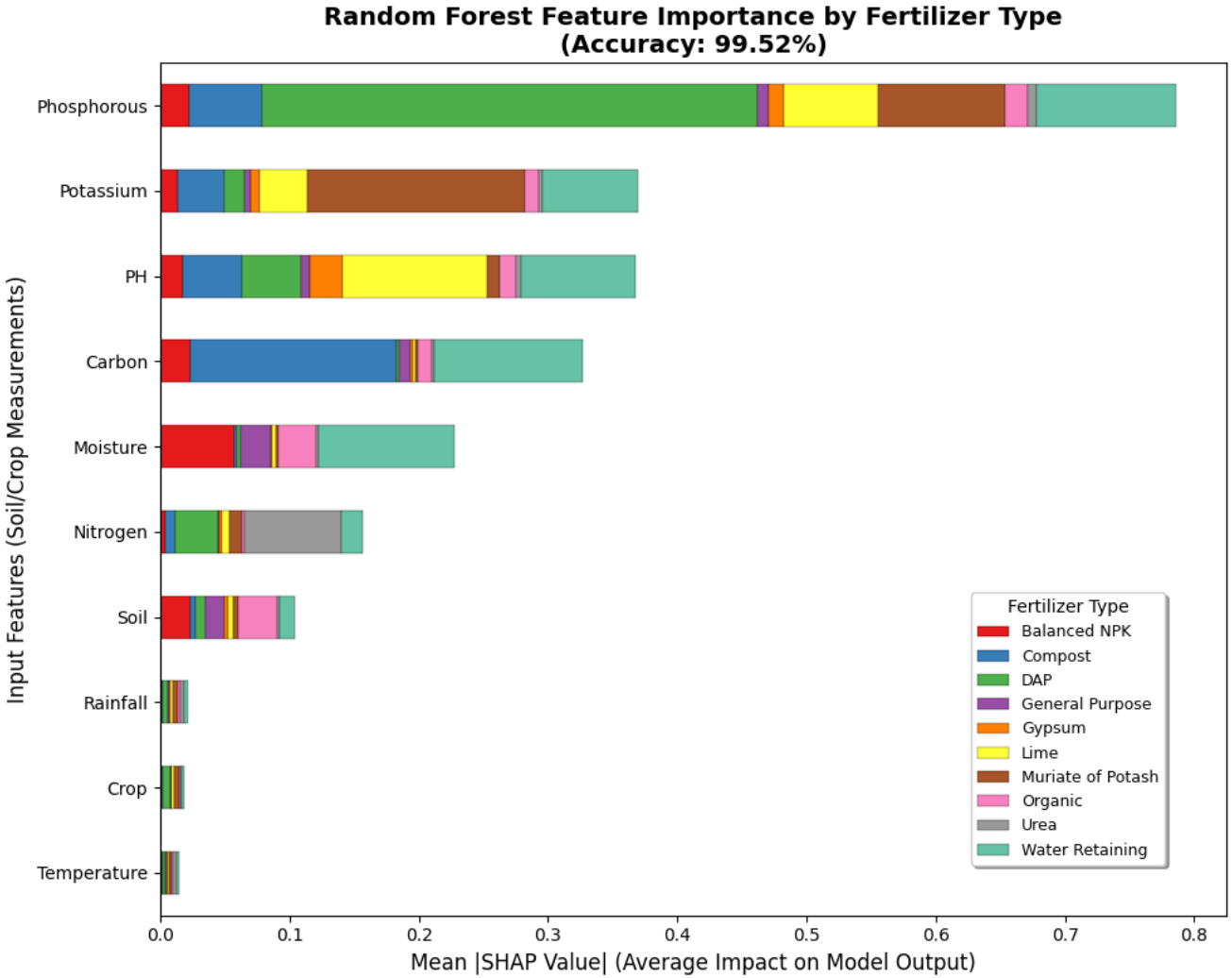


Figure 4: SHAP summary plot showing feature importance and impact on model predictions

Key Results

Model	Accuracy
Random Forest	0.9952 (99.52%)
XGBoost	0.9839 (98.39%)

Table 1: Model Performance Comparison on Test Set

- Random Forest outperformed XGBoost in prediction accuracy (99.52% vs 98.39%).

Practical and Policy-Relevant Insights

- Data-driven fertilizer use reduces misuse and costs.
- Precision agriculture improves soil sustainability.
- Supports informed decision-making for farmers.

Conclusion

- Machine learning effectively supports fertilizer recommendations.
- Random Forest demonstrated superior performance.
- Approach promotes sustainable agricultural practices.

Thank You